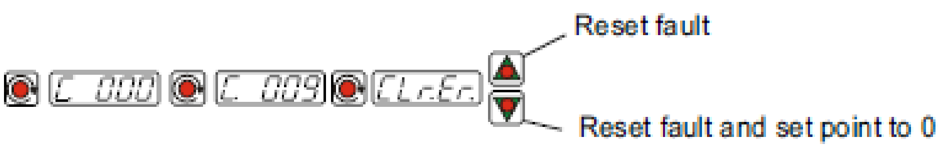
5 – FAULT MESSAGES

These messages show conditions that cause the controller to switch off!
To reset a fault either press the green button or go to C009, when at C009 press up arrow to reset fault, or down arrow to reset fault and change set point (A=0)

Error	Acc.	Sensor fault (only in closed loop) e.g. sensor defect or signal lost
Error	OC.	Over current fault, short-circuit (e.g. shorted turns in drive magnet)
Error	OL.	Overload fault (e.g. too much product on feeder)
Error	OU.	Over voltage fault (e.g. supply voltage too high; greater than 240V + 10%, or feedback from magnet)
Error	PEA.	Peak current limit exceeded (e.g. magnet air-gap too wide or a rapid frequency change)
Error	E.TP.	External temperature monitor (e.g. open temperature switch in magnet or no link between terminals 51, 52)
Error	E.L.I.	Internal program fault (not resettable - control unit must be returned to supplier)

Fault reset (Code 009 or green button)



PPM TECHNOLOGIES

6 – TROUBLESHOOTING

Problem	Possible Cause	Remedy	Procedure
Feeder does not vibrate	• Amplitude is set to zero • Incorrect frequency setting	• Increase amplitude by pressing [P] key twice in Menu C 000 and using arrow keys • Carry out resonant frequency search	Frequency search…. For controller with a feedback sensor connected • Empty feeder tray • Switch controller on • Adjust set point (A) to zero • Select code C038 • Enable Parameter ACC (set to 1) • Select Parameter AFS and press the top arrow key to start frequency search For controller without a feedback sensor • Empty feeder tray • Switch controller on • Set output frequency F in Code 038 to 100Hz • Adjust set point to a low value (approx. 30%) • Carefully increase or decrease F under Code 038 (direction depends on the feeder) • Resonance is reached when the amplitude is at maximum for a constant set point.
Feeder will not "settle" at resonance in AFS	• Feedback signal is too weak • Sensor air-gap is too wide	• Reduce power limit [P] in Menu 38 • Check sensor air-gap setting	
Feeder "hammers" when set point is high	• Feeder is operating too close to the resonant frequency • Coil air-gap is too small	• Adjust frequency • Reduce power limit [P] in Menu 38 • Check air-gap. (Caution - too wide an air-gap will increase current draw)	• Check that the air-gap setting is correct for the feeder - if necessary ask the manufacturer
Coil gets hot	• Frequency is set too low for the coil type • Air-gap is too wide	• Increase the frequency or use a different coil • Reduce the coil air-gap	• Adjust setting F under Code 038

6 – TROUBLESHOOTING (continued)

Problem	Possible Cause	Remedy	Procedure
"OFF" is displayed, Feeder does not run	• No enable signal • Coil thermal sensor "open circuit"	• Switch contacts or provide 24V enable signal • Check that link is fitted between terminals 5 & 6 • Check that link is fitted between terminals 51 & 52	• If the enable is not used then a link must be fitted between terminals 5 & 6 • If a 24V DC signal is used then the link must be removed
Feeder starts up slowly when enable is switched on, even though the soft start time is set to 0 (Occurs only in regulation mode)	• Maximum power limit [P] in Menu C 038 has not been set correctly • Circuit gain is set too low	• Adjust the maximum amplitude limit [P] in Menu C038 • Set PA (proportional characteristic or circuit gain)	• Check the set point value under Code 038. If it is 20% for example then the limit P can be reduced to 30%. After adjustment the set point range should then be 0 to 100% • In regulation mode the regulation gain of the electronic circuit must be tuned to the mechanical system. This is achieved by adjusting parameter PA under Code 038. If the feeder responds too slowly then the value must be increased to a level just below where the feeder oscillates ("hunts"). If the feeder "hunts" then the PA value must be reduced
Maximum amplitude is achieved with a very low set point value	• Maximum power limit [P] in Menu C 038 has not been set correctly	• Adjust the maximum amplitude limit [P] in Menu C038	• Increase the value of [P] in Menu C 038
ERROR - OL: Output power too high	• Coil power is too high	• Use a controller with a higher current rating	
	• Frequency is set too low	• Increase frequency	• Use parameter F under Code 038 Depending on the feeder characteristics the frequency can be set higher
	• Coil air-gap is too large	• Reduce air-gap	
	• Short-circuit	• Check wiring and coil	• To establish if a wiring short-circuit has occurred first remove connector for terminals 41, 42 & 43

6 – TROUBLESHOOTING (continued)

Problem		Possible Cause	Remedy	Procedure
ERROR - OC	Current too high	• Short-circuit on output • Coil faulty	• Check coil and wiring	• To establish if a wiring short-circuit has occurred first remove connector for terminals 41, 42 & 43
ERROR - OU	DC link voltage too high	• Mains voltage too high	• Check mains supply	
		• Back emf from coil (more likely at lower frequencies)	• Contact supplier	
ERROR - ACC	Sensor fault	• Sensor has failed	• Check sensor	• If the sensor is not used the parameter ACC must be set to 0 in Menu C 038
ERROR - EEP	Memory fault	• Component problem	• Refer to supplier	• Unit cannot be repaired on site (exchange and send back to manufacturer
ERROR - PLL ERROR - IOA	µPC fault warning	• High EMC interference	• Check the integrity of the earth bonding of the controller and connections to the feeder	• Feeder, sensor and enable input must be connected with screened cables • Sensor and enable input cables should not be routed in the same trunking as power cables • If this code is displayed it may be possible to reset the system by switching the mains supply off and then back on again • If the fault reoccurs refer to supplier
Original Controller settings have been lost		• On-site adjustment	• Recall conveyor settings	• ISet USPR to SAFE in Menu 210

If the above procedures do not solve the problem and it appears that the unit is faulty then please use the check sheet provided before returning the controller to your supplier.